

CEPA Logic PRO + AdBuster PRO

Intelligent Audio Comfort Protection

1. Introduction

CEPA Logic PRO and AdBuster PRO work together to deliver a fully automated, intelligent system for stabilizing TV audio. Their combined design eliminates loud commercials, sudden volume jumps, and uncomfortable listening moments — all without user interaction and without internet connectivity.

This system operates entirely offline, ensuring privacy, reliability, and instant response.

2. How the System Works

2.1 AdBuster PRO — Audio Event Detection

AdBuster PRO includes a local machine-learning model that identifies:

- Commercial segments
- Normal program audio
- Aggressive or overly loud fragments

The model runs on the device, without cloud processing or data transmission.

2.2 CEPA Logic PRO — Behaviour Engine

CEPA Logic PRO interprets the audio context and determines the correct system behaviour. It does not classify audio — it reacts to the signals provided by AdBuster PRO.

CEPA activates:

- Comfort-protection mode
- Transition smoothing
- Volume stabilization
- Spike-response logic
- Natural dynamics preservation

2.3 IR Execution Layer

Once CEPA decides on the appropriate action, AdBuster PRO sends precise IR commands to the TV:

- Lowering volume
- Holding volume steady
- Restoring normal levels

All actions are performed automatically and immediately.

3. Key Benefits

3.1 Effortless Listening Comfort

The system removes the need for manual volume adjustments by eliminating:

- Loud commercials
- Sudden audio peaks
- Abrupt transitions
- Inconsistent program levels

3.2 Fully Offline Operation

CEPA Logic PRO and AdBuster PRO operate locally:

- No cloud
- No streaming
- No data collection
- No privacy concerns

3.3 Precision and Reliability

Each module performs a dedicated role:

- **AdBuster PRO** detects audio events
- **CEPA Logic PRO** decides how to respond
- **AdBuster PRO (IR)** executes the action

This separation ensures stable, predictable behaviour in all conditions.

4. System Architecture Overview

AdBuster PRO → CEPA Logic PRO → IR Control

1. **Detection** Local ML model identifies commercials and high-intensity audio.
2. **Interpretation** CEPA Logic PRO evaluates the context and selects the correct behavioural response.
3. **Execution** IR commands adjust the TV volume with precision and speed.

5. Summary

CEPA Logic PRO and AdBuster PRO form a unified, intelligent audio-comfort system designed to protect the listener from disruptive volume changes. By combining real-time detection, behavioural interpretation, and precise IR control, the system delivers a smooth, stable, and enjoyable audio experience — fully automatic and fully offline.